25

Lecture 25

Network Evolution: a Grand Synthesis

تکامل شبکه: یک جمع‌بندی بزرگ

2G — 3G — 4G — **5G** — 6G

COURSE PART
Part V – Planes and outlook (Lectures 23–26)

ESTIMATED STUDY TIME
6–8 hours (3 h reading, 3–5 h exercises)

DIFFICULTY
Advanced

PREREQUISITES
Lectures 1–24

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Original lecture notes for the course *Advanced Communication Networks*. The lecture sequence follows the course syllabus (26 lectures, 2G to 6G). Equations use identical notation in the English and Persian editions. Companion edition: Persian (`lecture-25-fa.pdf`).

1 Learning objectives

After studying this lecture you should be able to:

- **LO1** **Trace** the evolution of radio access, core architecture, signalling, security and QoS from 2G to 5G.
- **LO2** **Map** network functions across generations (MSC/VLR, MME, AMF; HLR, HSS, UDM; GGSN, P-GW, UPF).
- **LO3** **Decompose** capacity growth into spectrum, spectral efficiency and densification.
- **LO4** **Quantify** long-term trends in peak rate, latency and cost per bit with simple growth models.
- **LO5** **Identify** the design principles that recur in every generation and will shape the next one.

2 Prerequisite knowledge

This lecture draws on the whole course: GSM, GPRS and the 2G/3G core (Lectures 1–4), LTE and the EPC (Lectures 5–13), and the 5G system, SDN, NFV and SBA (Lectures 14–24). No new protocol is introduced; the aim is to connect what you already know into one picture.

Key terms. The vocabulary of this lecture with its Persian equivalents:

Term	Persian	Meaning in this lecture
Generation	نسل	A family of standards meeting a set of ITU requirements (IMT-2000, IMT-Advanced, IMT-2020).
Flat architecture	معماری تخت	Fewer nodes on the user path; radio control in the base station.
All-IP	تمام‌IP	Voice and data carried as IP packets, no circuit-switched domain.
CP/UP separation	جداسازی صفحهٔ کنترل و کاربر	Independent scaling and placement of control and user plane.
Spectral efficiency	بازدهی طیفی	Throughput per hertz per cell (bit/s/Hz/cell).
Densification	متراکم‌سازی	More cells per unit area.
Area traffic capacity	ظرفیت ترافیکی ناحیه	Total throughput per unit area (bit/s/m^2).
Cost per bit	هزینه به ازای بیت	Network cost divided by carried traffic.
Softwarisation	نرم‌افزاری‌شدن	Network functions as software on general-purpose infrastructure.
Backward compatibility	سازگاری با گذشته	Interworking of a new generation with installed ones.

3 Introduction and motivation

Each generation of mobile networks was designed to fix the bottleneck of the previous one: 2G digitised voice and added security and roaming; 3G brought packet data at useful rates; 4G removed the circuit domain and flattened the network; 5G made the network programmable and service-specific. Seen one at a time, the generations look like a sequence of unrelated protocol stacks. Seen together, they show a small set of persistent ideas – separation of concerns, mobility anchors, subscriber databases, tunnelling, mutual authentication – implemented with ever more general technology [3], [7]. Recognising these ideas is what lets an engineer understand a new system quickly, and predict where it will go.

4 Fundamental concepts

4.1 Five dimensions of evolution

- **Radio access**: FDMA/TDMA (GSM), CDMA (UMTS), OFDMA with MIMO (LTE), flexible-numerology OFDM with massive MIMO and mmWave (NR).
- **Core**: circuit switching with MSC/VLR and SS7/MAP; an overlaid packet core (SGSN/GGSN); the all-IP EPC with Diameter and GTP; the service-based 5GC on cloud infrastructure.
- **Mobility**: hierarchical location and routing areas, anchors in the GGSN, P-GW and UPF; handover decisions moving from the BSC and RNC into the base station.
- **Security**: one-way GSM authentication (A3/A8) to mutual UMTS AKA, EPS-AKA with key hierarchy, and 5G-AKA/EAP-AKA′ with concealed identities (SUCI) and home control.
- **QoS**: from circuit guarantees through PDP contexts and EPS bearers to per-flow QoS with standardised 5QIs and slicing.

4.2 Flattening the path

The user-plane path has steadily shrunk (Figure 1). In GPRS and UMTS a packet passed through a base station, a radio controller (BSC/PCU or RNC), an SGSN and a GGSN. LTE moved radio control into the eNB and removed the controller; the S-GW and P-GW remained. In 5G a single UPF can serve as the whole core user plane, and it can be placed close to the radio.

4.3 Functions that survive

Names change, functions persist. The visitor location and mobility function (MSC/VLR, SGSN, MME, AMF), the subscriber database (HLR/AuC, HSS, UDM/AUSF), the mobility anchor and gateway to data networks (GGSN, P-GW, UPF with SMF) and policy control (PCRF, PCF) appear in every generation, each time with cleaner interfaces and better separation between control and user plane.

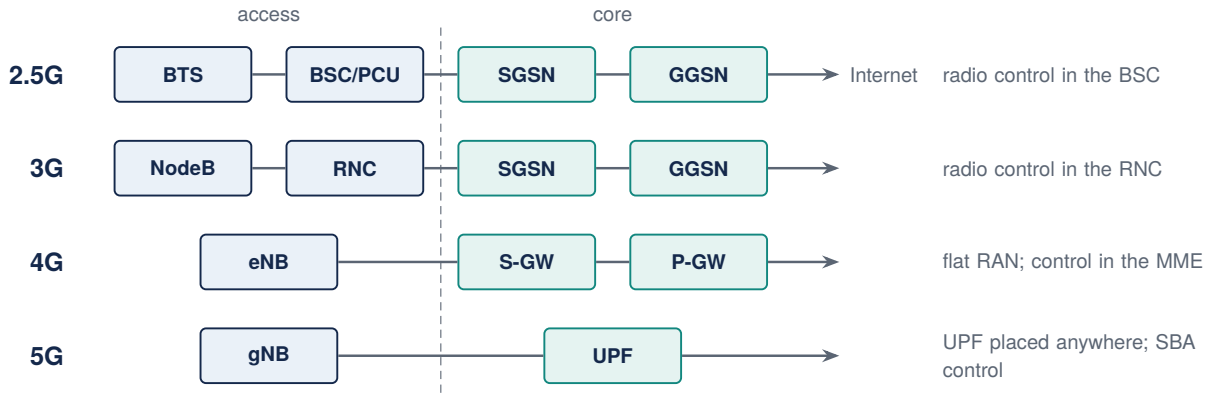


Figure 1. User-plane path of packet data from 2.5G to 5G (simplified).

5 Detailed technical explanation

5.1 From circuits to packets to services

2G was designed around a 64 kbit/s circuit per call. GPRS added a packet core in parallel, reusing the radio with dynamic time-slot allocation. UMTS kept two domains (CS and PS) with a common radio network. LTE dropped the CS domain: voice became an IMS service over VoLTE, with CS fallback and SRVCC for interworking. 5G keeps IMS for voice (VoNR) but restructures the core as services consumed over HTTP/2 – the network itself becomes a set of APIs.

5.2 Signalling protocols

SS7 and MAP connected the 2G/3G core; Diameter replaced MAP in the EPC (S6a, Gx, Rx); GTPv2-C carried session control; 5G uses REST-style service-based interfaces with OAuth-based authorisation, and PFCP for user-plane control. Each step moved towards protocols used in the wider IT industry, lowering development cost and enabling cloud deployment.

5.3 Security lessons

GSM authenticated only the subscriber, allowing false base stations; UMTS added network authentication and integrity protection; LTE separated keys per layer and per node; 5G encrypts the permanent identity (SUCI), gives the home network the final authentication decision and protects inter-operator signalling with the SEPP. Each fix answered an attack that became practical on the previous generation.

5.4 Coexistence

No generation replaces its predecessor at once. Networks run several in parallel for a decade or more, with interworking functions (N26 between AMF and MME, EN-DC in non-standalone 5G, CS fallback) and gradual spectrum refarming. Backward compatibility is a constant design constraint and a large part of the cost of every new system.

6 Mathematical models

6.1 Peak-rate growth

Model. An exponential trend $R(t) = R_0\, e^{k(t-t_0)}$ fitted through two points (t_0, R_0) and (t_1, R_1) has

$$k = \frac{\ln(R_1/R_0)}{t_1 - t_0}, \qquad T_2 = \frac{\ln 2}{k}. \tag{1}$$

Example. From GSM data (9.6 kbit/s, 1991) to the IMT-2020 requirement (20 Gbit/s, specified 2017–2018) [8]: $k = \ln(2.08 \times 10^6)/27 = 0.539$ per year, a factor 1.71 per year, doubling every 1.29 years (Figure 2).

Limitation. Peak rates are specification values under ideal conditions; user-experienced rates grow more slowly and depend on load and coverage.

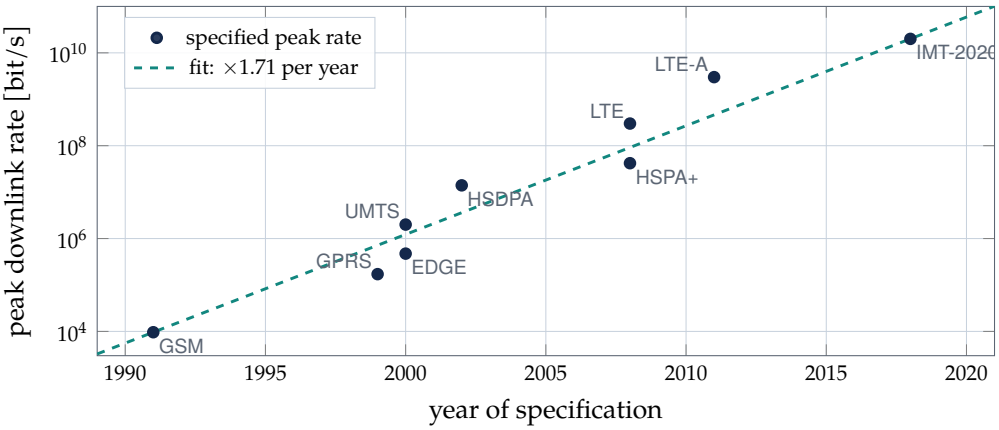


Figure 2. Specified downlink peak rates of 3GPP technologies and the exponential fit of Eq. (1).

6.2 Decomposing capacity

Model. The area traffic capacity is the product of cell density n (cells/km^2), bandwidth B and spectral efficiency η:

$$C_A = n\, B\, \eta, \tag{2}$$

so the growth between two deployments is the product of the three ratios.

Example. LTE baseline: $n = 4$, $B = 20$ MHz, $\eta = 1.5$ bit/s/Hz: $C_A = 120$ Mbit/s/km^2. 5G deployment: $n = 16$, $B = 100$ MHz, $\eta = 6$: $C_A = 9.6$ Gbit/s/km^2 – a factor $4 \times 5 \times 4 = 80$ (Figure 3). For comparison, IMT-2020 requires 10 Mbit/s/m^2 (10 Tbit/s/km^2) in the indoor hotspot scenario [8].

6.3 Latency along the path

Model. The one-way user-plane latency through h nodes is

$$T = \sum_{i=1}^{h} \left(t_{\mathrm{proc},i} + \frac{d_i}{v} \right), \tag{3}$$

with processing time $t_{\mathrm{proc},i}$, link length d_i and propagation speed $v \approx 2 \times 10^8$ m/s in fibre.

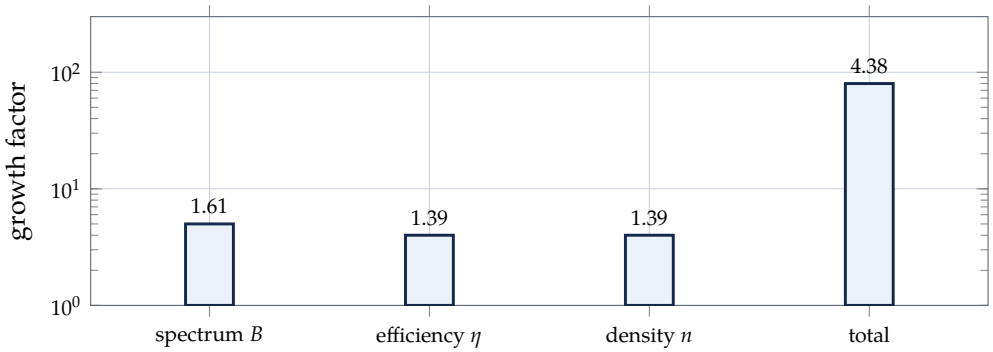


Figure 3. Area-capacity growth from the LTE baseline to the 5G deployment of Section 6.2.

Example. Four nodes with 0.5 ms each and 400 km of fibre (2 ms) give 4 ms; two nodes (gNB and edge UPF) with 0.25 ms each and 20 km give 0.6 ms. Flattening and edge placement together remove most of the core latency.

6.4 Cost per bit

Model. If traffic grows by a factor $1+g$ per year and revenue by $1+r$, the cost per bit must fall by $(1+r)/(1+g)$ per year to keep margins constant; it halves every

$$T_{1/2} = \frac{\ln 2}{\ln(1+g) - \ln(1+r)} \text{ years.} \tag{4}$$

Example. With $g = 30\,\%$ and flat revenue, $T_{1/2} = 2.64$ years. This pressure, more than peak rate, explains the industry's move to commodity hardware, automation and software-defined networks.

7 Network architecture and diagrams

Figure 1 shows the user path; the control side evolved in parallel from MSC/VLR and SGSN to the MME and then to the AMF and SMF of the service-based core. Table 1 maps the functions across generations.

Table 1. Corresponding functions across generations.

Function	2G/3G	4G (EPC)	5G (5GC)
Mobility and access control	MSC/VLR, SGSN	MME	AMF
Subscriber data and authentication	HLR/AuC	HSS	UDM, AUSF
Session control	SGSN, GGSN	MME, S-GW/P-GW control	SMF
User-plane anchor	GGSN	P-GW (S-GW)	UPF
Policy	– (static)	PCRF	PCF
Radio control	BSC, RNC	eNB	gNB (CU-CP)

8 Step-by-step procedures

Procedure 1 • Analysing a new mobile system

1. Identify the subscriber database, the mobility function, the session function and the user-plane anchor.
2. Draw the user path and count the nodes and tunnels (Eq. (3)).
3. Identify the authentication method and which identities travel in clear.
4. Identify the QoS unit (circuit, bearer, flow) and where it is enforced.
5. Locate the interworking with the previous generation.

Procedure 2 • Planning a capacity increase

1. Measure current traffic and forecast growth.
2. Compute the required factor and split it into spectrum, efficiency and density (Eq. (2)).
3. Price each lever (licences, new equipment, sites and backhaul).
4. Choose the cheapest combination and check the cost-per-bit target (Eq. (4)).

9 Worked examples

Worked example 1 • Densification needed

Problem. A hotspot must carry 10 Tbit/s/km^2 with $B = 400$ MHz and $\eta = 10$ bit/s/Hz. How many cells per km^2 are needed?
Solution. $n = 10^{13}/(4 \times 10^8 \times 10) = 2500$ cells/km^2, one cell per 400 m^2 – an inter-site distance of about 20 m, i.e. indoor small cells.

Worked example 2 • Extrapolating the trend

Problem. If the fit of Section 6.1 continued, when would peak rates reach 1 Tbit/s, a figure mentioned in 6G visions [10]?
Solution. $\ln(10^{12}/2 \times 10^{10})/0.539 = 7.3$ years after 2018, i.e. around 2025. That has not happened, showing that the trend is not a law: further growth needs wider bandwidths that exist only above 100 GHz (Lecture 26).

Worked example 3 • Cost pressure

Problem. Traffic grows 25 % per year and revenue 2 %. By how much must cost per bit fall each year, and when does it halve?
Solution. $1.02/1.25 = 0.816$: an 18.4 % fall per year; $T_{1/2} = \ln 2/(\ln 1.25 - \ln 1.02) = 3.41$ years.

10 Technology comparison

Table 2. Generations at a glance.

	2G (GSM)	3G (UMTS)	4G (LTE)	5G (NR)
Multiple access	TDMA/FDMA	WCDMA	OFDMA / SC-FDMA	OFDMA, flexible numerology
Core	CS (+ GPRS)	CS + PS	All-IP EPC	Service-based 5GC
Signalling	SS7/MAP	SS7/MAP	Diameter, GTPv2-C	HTTP/2 SBI, PFCP
Authentication	One-way	Mutual AKA	EPS-AKA	5G-AKA, EAP-AKA′, SUCI
QoS unit	Circuit	PDP context	EPS bearer	QoS flow, slice
Voice	CS	CS	VoLTE (IMS)	VoNR (IMS)

Same ideas, better tools. Every generation keeps the same functional roles but rebuilds them with more general technology: from dedicated switches to IP routers to cloud software.

11 Key takeaways

Key takeaways

- Functions persist across generations; their names, protocols and placement change.
- The user path flattened from four nodes (GPRS/UMTS) to one UPF in 5G.
- Capacity $C_A = nB\eta$: growth comes from spectrum, efficiency and densification together.
- Peak rate grew about 1.7 times per year from 1991 to 2018, but trends are not laws.
- Cost per bit must halve every $\ln 2 / (\ln(1+g) - \ln(1+r))$ years – the driver of softwarisation.

12 Exercises

Exercise 1 • Level 1 – conceptual • Mapping

Name the 4G and 5G counterparts of: (a) VLR; (b) HLR; (c) GGSN; (d) RNC.

Exercise 2 • Level 1 – conceptual • Why packets?

Give two reasons why mobile networks moved from circuit to packet switching for all traffic.

Exercise 3 • Level 2 – numerical • Growth rate

LTE Rel-8 offered 300 Mbit/s (2008) and LTE-Advanced 3 Gbit/s (2011). Compute k and the doubling time for this interval and compare with Section 6.1.

Exercise 4 • Level 2 – numerical • Capacity levers

An operator needs 20 times more area capacity. Spectrum can grow by 2.5 and efficiency by 2. What densification is needed?

Exercise 5 • Level 3 – analytical • Path latency

Compare with Eq. (3): (a) UMTS path with NodeB, RNC, SGSN and GGSN (1 ms each, 300 km fibre); (b) 5G with gNB and a UPF 30 km away (0.3 ms each).

Exercise 6 • Level 3 – analytical • The next bottleneck

Using the five dimensions of evolution, argue which bottleneck of 5G a sixth generation is most likely to address, with a quantitative argument.

Solutions

Solution to Exercise 1

(a) MME / AMF; (b) HSS / UDM (with AUSF); (c) P-GW / UPF with SMF; (d) none as a separate node – eNB / gNB (CU-CP).

Solution to Exercise 2

Statistical multiplexing uses capacity far better for bursty traffic, and a single IP network for voice and data lowers cost and allows Internet services to be offered directly.

Solution to Exercise 3

$k = \ln 10/3 = 0.768$ per year, $T_2 = 0.90$ years – faster than the long-term 1.29 years, because carrier aggregation and MIMO added bandwidth and layers in a single step.

Solution to Exercise 4

$20/(2.5 \times 2) = 4$: four times as many cells per unit area.

Solution to Exercise 5

(a) $4 \times 1 + 300\,\text{km}/(2 \times 10^5\,\text{km/s}) = 4 + 1.5 = 5.5\,\text{ms}$. (b) $2 \times 0.3 + 0.15 = 0.75\,\text{ms}$.

Solution to Exercise 6

Example argument: the peak-rate trend stalled because bandwidth below 100 GHz is exhausted; $C = B \log_2(1 + \text{SNR})$ grows linearly in B but only logarithmically in SNR, so the next tenfold step needs tens of gigahertz of spectrum (sub-THz), plus new ways to reduce cost per bit (AI-driven automation) and to cover the unconnected (non-terrestrial networks).

13 Further reading

ITU-R M.2083 [7] and M.2410 [8] define the IMT-2020 vision and requirements; TS 23.002 [2], TS 23.401 [1] and TS 23.501 [3] describe the architectures of the successive generations. Eberspächer et al. [5], Holma and Toskala [6], Sesia et al. [11] and Dahlman et al. [4] cover GSM, UMTS, LTE and NR; Rommer et al. [9] cover the 5G core.

References

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